

DYLAN OESCH

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EDUCATION

Western University

Expected May 2029

Bachelor of Engineering Science in Software Engineering / Co-op Program

London, Ontario

- **Relevant Coursework:** Algorithms and Data Structures, Software Design, Software Construction, Digital Logic Systems.

PROJECTS

ARGUS Local AI Assistant

Summer 2026 – Present

Python, FastAPI, JavaScript, Linux, Artificial Intelligence

- Designed and built a local voice-controlled AI assistant integrating speech recognition, local language-model inference, text-to-speech, and Google services through a Python and FastAPI backend to provide hands-free access to personal tools and information.
- Deployed the system on a Linux-based Lenovo ThinkCentre with a 7-inch touchscreen, browser-based interface, automatic kiosk startup, and custom system services for reliable standalone operation.
- Designed and assembled a multi-part 3D-printed enclosure, iterating on component fit, tolerances, fastening methods, print orientation, and hardware accessibility to produce a functional first prototype.

Two-Player Bomb Defusal Game

November 2025

Arduino Uno, Embedded Programming, Circuit Design

- Developed a functional two-player physical number-guessing game using an Arduino Uno, five push-button inputs, a timed bomb mechanic, and a four-state gameplay system to deliver interactive turn-based gameplay.
- Designed the circuit schematic and programmed the game-control logic to manage player transitions, button inputs, timing, answer detection, and end-game states, enabling reliable operation of the complete breadboard prototype.

EXPERIENCE

Calhoun Super Structures

May 2026 – August 2026

Manufacturing/Industrial Engineering Co-op Student / Software & Automation

Goderich, Ontario

- Developed a Python desktop middleware application to automate the transfer of SolidWorks assembly data into validated Epicor parts, revisions, bills of materials, and manufacturing operations, reducing upload time by approximately 85%.
- Built and deployed an Epicor-to-ClickUp project visibility system using Power Automate, Excel, and application programming interfaces (APIs) to manage approximately 500 project records, reducing update time by 80%.

Courtney Custom Masonry

June 2022 – September 2025

Stonemason

Varna, Ontario

- Constructed brick and stone masonry across approximately 15–20 residential homes, independently laying stone to site specifications and quality requirements.
- Operated telehandlers and other job-site machinery independently to transport materials and maintain continuous supply to masonry crews throughout construction.

Domino's

February 2023 – September 2025

Manager

Goderich, Ontario

- Led shifts of approximately 10 employees within a 30-person store, coordinating daily operations, till balancing, deposits, and closing for roughly \$4,000 in daily sales.
- Trained 15+ employees across front-counter, oven, and kitchen operations, preparing staff to work effectively across multiple store roles.

ENGINEERING ACTIVITIES

Western Formula Racing

September 2024 – May 2025

General Member

London, Ontario

- Created and modified SolidWorks CAD models for automotive components during team design activities, gaining practical experience with engineering design workflows.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, HTML, CSS

Hardware and Design: Arduino, SolidWorks, Fusion 360, AutoCAD, Onshape, Embedded Prototyping, 3D Printing

Systems and APIs: Linux, FastAPI, REST APIs, Google APIs, OAuth 2.0, Git, GitHub

Automation and Data: Power Automate, Power BI, Power Query, DAX, Excel, Epicor, ClickUp